Database Management System LAB
Spring Semester 2024-25
Lab Assignment 2

Name – Soham Chakraborty
Roll No – 22CS02002

Ans – 1)

```
mysql> create database office;
Query OK, 1 row affected (0.01 sec)

mysql> use office;
Database changed
mysql> create table Employee(
    -> Eno INT PRIMARY KEY,
    -> Ename Varchar(50),
    -> DeptNo INT,
    -> Salary INT,
    -> Sex Char(1),
    -> DoB DATE
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> create table Department(
    -> Dno int PRIMARY KEY,
    -> Dname varchar(50),
    -> Dloc varchar(20),
    -> Dmgrno int
    -> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> alter table Employee
    -> Add Foreign Key (DeptNo)
    -> References Department(Dno);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> alter table Department
    -> Add Foreign Key (Dmgrno)
    -> References Employee(Eno);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> alter table Employee
    -> Add Constraint check_salary
    -> Check (Salary>=10000);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Ans – 2) Since ‘DeptNo’ of Employee Table requires ‘Dno’ of Department Table to be defined first, it is set to NULL for now and will be declared later on after Department Table is Created.

```
mysql> Insert into Employee (Eno,Ename,Salary,Sex,DoB)
    -> Values (1,'Soham',10000,'M','2024-01-20');
Query OK, 1 row affected (0.01 sec)

mysql> Insert into Employee (Eno,Ename,Salary,Sex,DoB)
    -> Values
    -> (2,'Alan',150000,'M','2000-01-01'),
    -> (3,'Thor',50000,'M','1990-01-01'),
    -> (4,'Rose',20000,'F','2004-01-01'),
    -> (5,'Thomas',200000,'M','1947-08-15'),
    -> (6,'Andrew',80000,'M','2001-09-11'),
    -> (7,'Emily',125000,'F','1980-01-01'),
    -> (8,'Tommy',200000,'M','1975-01-01'),
    -> (9,'Twilight',75000,'M','1995-01-01'),
    -> (10,'Harry',100000,'M','2001-11-16');
Query OK, 9 rows affected (0.06 sec)
Records: 9  Duplicates: 0  Warnings: 0

mysql> select * from Employee;
+-----+----------+--------+--------+------+------------+
| Eno | Ename    | DeptNo | Salary | Sex  | DoB        |
+-----+----------+--------+--------+------+------------+
|   1 | Soham    |   NULL |  10000 | M    | 2024-01-20 |
|   2 | Alan     |   NULL | 150000 | M    | 2000-01-01 |
|   3 | Thor     |   NULL |  50000 | M    | 1990-01-01 |
|   4 | Rose     |   NULL |  20000 | F    | 2004-01-01 |
|   5 | Thomas   |   NULL | 200000 | M    | 1947-08-15 |
|   6 | Andrew   |   NULL |  80000 | M    | 2001-09-11 |
|   7 | Emily    |   NULL | 125000 | F    | 1980-01-01 |
|   8 | Tommy    |   NULL | 200000 | M    | 1975-01-01 |
|   9 | Twilight |   NULL |  75000 | M    | 1995-01-01 |
|  10 | Harry    |   NULL | 100000 | M    | 2001-11-16 |
+-----+----------+--------+--------+------+------------+
10 rows in set (0.00 sec)
```

```
mysql> Insert into Department
    -> Values
    -> (1,'Software Developer','Bangalore',8),
    -> (2,'Human Resource','Delhi',4),
    -> (3,'Backend','Bombay',10);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Department;
+-----+--------------------+-----------+--------+
| Dno | Dname              | Dloc      | Dmgrno |
+-----+--------------------+-----------+--------+
|   1 | Software Developer | Bangalore |      8 |
|   2 | Human Resource     | Delhi     |      4 |
|   3 | Backend            | Bombay    |     10 |
+-----+--------------------+-----------+--------+
3 rows in set (0.00 sec)
```

Now after Dno (DeptNo) has been defined we go back to Employee Table and update the (DeptNo) values from NULL.

```
mysql> update Employee Set DeptNo = 2 Where Eno = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 1 Where Eno = 2;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 2 Where Eno = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 2 Where Eno = 4;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo =1 Where Eno = 5;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 3 Where Eno = 6;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 2 Where Eno = 7;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 1 Where Eno = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 3 Where Eno = 9;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Employee Set DeptNo = 3 Where Eno = 10;
Query OK, 1 row affected (0.07 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Updated Employee Table

```
mysql> select * from Employee;
+-----+----------+--------+--------+------+------------+
| Eno | Ename    | DeptNo | Salary | Sex  | DoB        |
+-----+----------+--------+--------+------+------------+
|   1 | Soham    |      2 |  10000 | M    | 2024-01-20 |
|   2 | Alan     |      1 | 150000 | M    | 2000-01-01 |
|   3 | Thor     |      2 |  50000 | M    | 1990-01-01 |
|   4 | Rose     |      2 |  20000 | F    | 2004-01-01 |
|   5 | Thomas   |      1 | 200000 | M    | 1947-08-15 |
|   6 | Andrew   |      3 |  80000 | M    | 2001-09-11 |
|   7 | Emily    |      2 | 125000 | F    | 1980-01-01 |
|   8 | Tommy    |      1 | 200000 | M    | 1975-01-01 |
|   9 | Twilight |      3 |  75000 | M    | 1995-01-01 |
|  10 | Harry    |      3 | 100000 | M    | 2001-11-16 |
+-----+----------+--------+--------+------+------------+
10 rows in set (0.00 sec)
```

Ans – 3)

(a)

```
mysql> select * from Employee where DeptNo = 1;
+-----+--------+--------+--------+------+------------+
| Eno | Ename  | DeptNo | Salary | Sex  | DoB        |
+-----+--------+--------+--------+------+------------+
|   2 | Alan   |      1 | 150000 | M    | 2000-01-01 |
|   5 | Thomas |      1 | 200000 | M    | 1947-08-15 |
|   8 | Tommy  |      1 | 200000 | M    | 1975-01-01 |
+-----+--------+--------+--------+------+------------+
3 rows in set (0.00 sec)
```

(b)

```
mysql> select * from Employee where TimeStampDiff(YEAR,DoB,CURDATE()) < 30;
+-----+--------+--------+--------+------+------------+
| Eno | Ename  | DeptNo | Salary | Sex  | DoB        |
+-----+--------+--------+--------+------+------------+
|   1 | Soham  |      2 |  10000 | M    | 2024-01-20 |
|   2 | Alan   |      1 | 150000 | M    | 2000-01-01 |
|   4 | Rose   |      2 |  20000 | F    | 2004-01-01 |
|   6 | Andrew |      3 |  80000 | M    | 2001-09-11 |
|  10 | Harry  |      3 | 100000 | M    | 2001-11-16 |
+-----+--------+--------+--------+------+------------+
5 rows in set (0.00 sec)
```

(c)

```
mysql> Select Ename,Salary from Employee
    -> Join Department on Employee.Eno = Department.Dmgrno
    -> where Employee.Eno = Department.Dmgrno;
+-------+--------+
| Ename | Salary |
+-------+--------+
| Rose  |  20000 |
| Tommy | 200000 |
| Harry | 100000 |
+-------+--------+
3 rows in set (0.00 sec)
```

(d)

```
mysql> Select d.Dname,e.Ename from Employee e
    -> Join Department d on e.Eno = d.Dmgrno
    -> where e.Eno = d.Dmgrno;
+--------------------+-------+
| Dname              | Ename |
+--------------------+-------+
| Software Developer | Tommy |
| Human Resource     | Rose  |
| Backend            | Harry |
+--------------------+-------+
3 rows in set (0.00 sec)
```

(e)

```
mysql> Select Ename from Employee where Salary > 100000 and Deptno = 1;
+--------+
| Ename  |
+--------+
| Alan   |
| Thomas |
| Tommy  |
+--------+
3 rows in set (0.00 sec)
```

(f)

```
mysql> select Ename from Employee where Salary = (select max(Salary) from Employee);
+--------+
| Ename  |
+--------+
| Thomas |
| Tommy  |
+--------+
2 rows in set (0.00 sec)
```

(g)

```
mysql> update Employee
    -> set Salary = 1.1*Salary
    -> where DeptNo = 2;
Query OK, 4 rows affected (0.00 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> select * from Employee;
+-----+----------+--------+--------+------+------------+
| Eno | Ename    | DeptNo | Salary | Sex  | DoB        |
+-----+----------+--------+--------+------+------------+
|   1 | Soham    |      2 |  11000 | M    | 2024-01-20 |
|   2 | Alan     |      1 | 150000 | M    | 2000-01-01 |
|   3 | Thor     |      2 |  55000 | M    | 1990-01-01 |
|   4 | Rose     |      2 |  22000 | F    | 2004-01-01 |
|   5 | Thomas   |      1 | 200000 | M    | 1947-08-15 |
|   6 | Andrew   |      3 |  80000 | M    | 2001-09-11 |
|   7 | Emily    |      2 | 137500 | F    | 1980-01-01 |
|   8 | Tommy    |      1 | 200000 | M    | 1975-01-01 |
|   9 | Twilight |      3 |  75000 | M    | 1995-01-01 |
|  10 | Harry    |      3 | 100000 | M    | 2001-11-16 |
+-----+----------+--------+--------+------+------------+
10 rows in set (0.00 sec)
```

```
mysql> select Ename from Employee where Ename LIKE 'Th%';
+--------+
| Ename  |
+--------+
| Thor   |
| Thomas |
+--------+
2 rows in set (0.00 sec)
```

(i) Here, location of Department is chosen as ‘Delhi’ instead of ‘Bangalore’ because there were no female Employees in ‘Bangalore’.

```
mysql> Select sum(Salary) from Employee e Join Department d on e.DeptNo = d.Dno where d.Dloc = 'Delhi' and e.Sex = 'F';
+------------+
| sum(Salary) |
+------------+
|     159500 |
+------------+
1 row in set (0.00 sec)
```